

Precept 5 Outline

ECO 202 TAs

October 8, 2026

1 Weekly Check In

1. Exam 2 is on Monday! It covers primarily lectures 6 through 9. A couple of heads up:
 - No notes (no “cheat sheet”), no electronics
 - Format adjusted slightly – there will be room for scratch work, but we will only grade material in the designated spaces of the questions.
2. Any questions about the class, comments?
3. Reminders about resources:
 - Come to office hours for any preceptor, as much as you want.
 - Dexin: Tuesdays, 4-5pm, tables outside of JRR A19
 - Emily: Thursdays, 4:30-5:30pm, JRR A74
 - Ryan: Fridays, 2-3pm, JRR A74
 - Come to office hours of the professor: Fridays, 3pm-4pm, JRR 278
 - Check out the McGraw Center:
 - Individual tutoring: [Link to page](#)
 - Group study with a tutor: [Link to page](#)
 - Reach out to us by email if you’ve tried the above but are seeking support beyond these things.

2 Precept practice problems

1. (Practice 8, Question 4)

A procurement team uses a fair random mechanism to select exactly two of five suppliers, labeled 1 , 2 , 3 , 4 , 5 , for a documentation audit. The mechanism makes every unordered pair equally likely.

- (a) List all outcomes in Ω , determine its size, and explain why order must not create two copies of the same selected pair.

Solution. The sample space is

$$\Omega = \{12, 13, 14, 15, 23, 24, 25, 34, 35, 45\}.$$

For example, 12 and 21 describe the same selected suppliers and must not be counted twice. The mechanism explicitly makes the ten unordered pairs equally likely, so each has probability $1/10$.

- (b) Let S be the event that supplier 1 is selected. Let G be the event that at least one of suppliers 4 and 5 is selected. List the outcomes in S , G , and $S \cap G$.

Solution. The events are

$$S = \{12, 13, 14, 15\},$$

$$G = \{14, 15, 24, 25, 34, 35, 45\},$$

$$S \cap G = \{14, 15\}.$$

- (c) Calculate $P(S)$, $P(G)$, and $P(S \cap G)$ by counting outcomes.

Solution. $\mathbb{P}(S) = \frac{4}{10} = 0.4$, $\mathbb{P}(G) = \frac{7}{10} = 0.7$, $\mathbb{P}(S \cap G) = \frac{2}{10} = 0.2$.

- (d) Calculate $P(S \cup G)$ in two ways: directly from the sample space and with the addition rule.

Solution. Directly, the union contains every outcome except 23, so

$$\mathbb{P}(S \cup G) = \frac{9}{10} = 0.9.$$

The addition rule gives the same result:

$$\mathbb{P}(S \cup G) = 0.4 + 0.7 - 0.2 = 0.9.$$

- (e) Calculate the probability of neither S nor G and the probability of exactly one of S and G. Use the exhaustive event cells to check that your probabilities sum to one.

Solution. Neither event occurs only for 23, so its probability is $1/10 = 0.1$. Exactly one event occurs in the disjoint cells

$$S \cap G^c = \{12, 13\}$$

and

$$S^c \cap G = \{24, 25, 34, 35, 45\}.$$

Thus

$$\mathbb{P}(\text{exactly one}) = \frac{2+5}{10} = \frac{7}{10} = 0.7.$$

The four exhaustive-cell probabilities are 0.2 for both, 0.2 for S only, 0.5 for G only, and 0.1 for neither; they sum to one.

2. (Practice 9, Question 5)

A scholarship program applies an initial signal to 2,000 applications. Ten percent of the applications meet a sharply defined readiness benchmark. The signal is positive for 80% of benchmark-ready applications and for 20% of applications that do not meet the benchmark.

- (a) Define events for benchmark readiness and a positive signal, then construct the complete natural-frequency table.

Solution. Ready = R and positive signal = G .

Readiness	Negative signal (G^c)	Positive signal (G)	Total
Not ready (R^c)	1,440	360	1,800
Ready (R)	40	160	200
Total	1,480	520	2,000

- (b) Calculate the marginal probability of a positive signal using total probability.

Solution. Total probability gives

$$\mathbb{P}(G) = \mathbb{P}(G | R)\mathbb{P}(R) + \mathbb{P}(G | R^c)\mathbb{P}(R^c) = 0.80(0.10) + 0.20(0.90) = 0.26.$$

- (c) Calculate the probability that an application meets the benchmark given a positive signal, first from counts and then using Bayes' rule.

Solution. From the table,

$$\mathbb{P}(R | G) = \frac{160}{520} \approx 0.3077.$$

Bayes' rule agrees:

$$\mathbb{P}(R | G) = \frac{0.80(0.10)}{0.80(0.10) + 0.20(0.90)} = \frac{0.08}{0.26} \approx 0.3077.$$

- (d) Determine whether readiness and the signal are independent.

Solution. Readiness and the signal are not independent because $\mathbb{P}(G | R) = 0.80 \neq 0.26 = \mathbb{P}(G)$.

- (e) Predict how the posterior changes if the false-positive rate falls while the base rate and sensitivity stay fixed.

Solution. If the false-positive rate falls while sensitivity and the base rate stay fixed, fewer not-ready applications enter the positive-signal column, so $\mathbb{P}(R | G)$ rises.

3. (Previous ECO 202 HW Question)

In a certain population, 1% of individuals have a particular disease. A medical test is developed to detect this disease with the following characteristics:

- True Positive Rate: If a person has the disease, the test correctly identifies it 95% of the time.
- True Negative Rate: If a person does not have the disease, the test correctly indicates they are disease-free 95% of the time.

Questions:

- (a) If a randomly selected individual from the population tests positive, what is the probability that they actually have the disease?

Solution. $P(D|P) = \frac{P(P|D)P(D)}{P(P|D)P(D)+P(P|D^c)P(D^c)} = \frac{(.95)(.01)}{(.95)(.01)+(.05)(.99)} = 0.161$.

- (b) Explain why the probability calculated in part (a) is different from the test's accuracy of 95%.

Solution. Accuracy gives probability of positive test conditional on having the disease; part (a) asks for the reverse. Confusing these two or generally ignoring the impact of the low $P(D)$ on the above calculation is called the base-rate fallacy. What's going on is that there are two ways to test positive: by having the disease and having a true positive, or by not having the disease and having a false positive. Due to the extremely low base rate of the disease, the second channel is overall much more likely.

- (c) Suppose the prevalence of the disease in the population increases to 50%. Recalculate the probability that a person who tests positive actually has the disease.

Solution. $Pr(D|P) = \frac{(.95)(.5)}{(.95)(.5)+(.05)(.5)} = .95$ using the same logic as part (a).

- (d) **(Challenge/Bonus)** For the prevalence of 1%, if the same individual takes the test twice and tests positive both times, what is the probability that they actually have the disease? Assume that for a subject with a given disease status, the errors in the tests are independent.

Solution. $Pr(D|P^2) = \frac{Pr(P^2|D)Pr(D)}{Pr(P^2|D)Pr(D)+Pr(P^2|D^c)Pr(D^c)} = \frac{(.95)^2(.01)}{(.95)^2(.01)+(.05)^2(.99)} = 0.785$

3 Extra practice problems

1. In the United States, male heights (in inches) are approximately normally distributed with mean 69.1 and standard deviation 2.9. Female heights are approximately normally distributed with mean 63.7 and standard deviation 2.7. Suppose a randomly selected person is observed to have a height between 64.0 and 65.5 inches. What is the approximate probability that the person is female?

Solution. Assuming males and females are equally likely, let $H = 64 \leq \text{height} \leq 65.5$. Then $P(H|F) = \Phi((65.5 - 63.7)/2.7) - \Phi((64 - 63.7)/2.7) \approx 0.203$ and $P(H|M) = \Phi((65.5 - 69.1)/2.9) - \Phi((64 - 69.1)/2.9) \approx 0.068$. Thus, by Bayes' rule, $P(F|H) = \frac{0.203(0.5)}{0.203(0.5) + 0.068(0.5)} = \boxed{0.750 \text{ (approximately)}}$.

2. Show that if the events A, B are independent, then so are A, B^c .

Solution. Since $A = (A \cap B) \cup (A \cap B^c)$ disjointly, $P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)[1 - P(B)] = P(A)P(B^c)$. Thus $\boxed{A \text{ and } B^c \text{ are independent}}$.

3. Let A, B be events, and suppose $P(B) > 0$. Use the three axioms of probability to show the complement rule for conditional events, that is $P(A^c|B) = 1 - P(A|B)$.

Solution. Since $B = (A \cap B) \cup (A^c \cap B)$ disjointly, $P(B) = P(A \cap B) + P(A^c \cap B)$. Hence $P(A^c|B) = \frac{P(A^c \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)} = 1 - P(A|B)$.

4. Suppose $P(B) = 0.4$, $P(A|B^c) = 0.7$, and $P(A \cap B) = 0.1$. What is $P(B|A)$?

Solution. $P(A \cap B^c) = P(A|B^c)P(B^c) = 0.7(0.6) = 0.42$, so $P(A) = 0.1 + 0.42 = 0.52$. Therefore $P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.1}{0.52} = \boxed{\frac{5}{26} \approx 0.192}$.

5. Suppose the weight distribution of fish caught by a fishing trawler is approximately normal with mean 3.5 pounds and standard deviation 1.5 pounds. The processing facility on the trawler first sorts the fish into two categories by weight, with heavy fish being those over 5 pounds, and the remaining fish are considered light. Of the light fish, 25% are discarded, and of the heavy fish, 7% are discarded. Consider a randomly chosen discarded fish. What is the (approximate) probability that it is a heavy fish?

Solution. Let H = heavy and D = discarded. Since $P(H) = P(Z > (5 - 3.5)/1.5) = P(Z > 1) \approx 0.1587$, we have $P(H^c) \approx 0.8413$. Thus $P(H|D) = \frac{P(D|H)P(H)}{P(D|H)P(H) + P(D|H^c)P(H^c)} = \frac{0.07(0.1587)}{0.07(0.1587) + 0.25(0.8413)} = \boxed{0.050 \text{ (approximately)}}$.

6. Let A, B, C be events that all occur with positive probability. Suppose A, B are disjoint, A is independent of C , and B is independent of C . Further, $P(A) = 0.3$, $P(B \cap C) = 0.1$, and $P(C) = 0.5$.

(a) Is A independent of B ? Explain.

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Solution. No. Since A and B are disjoint, $P(A \cap B) = 0$, while $P(A)P(B) > 0$ because both occur with positive probability. Thus $\boxed{A \text{ and } B \text{ are not independent}}$.

(b) What is the probability that at least one of the three events happens?

Solution. Since B and C are independent, $0.1 = P(B \cap C) = P(B)(0.5)$, so $P(B) = 0.2$. Also $P(A \cap C) = P(A)P(C) = 0.3(0.5) = 0.15$. Because $A \cap B = \emptyset$, $P(A \cap B \cap C) = 0$. Therefore $P(A \cup B \cup C) = 0.3 + 0.2 + 0.5 - 0 - 0.15 - 0.1 + 0 = \boxed{0.75}$.

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7. Let A, B be events. Suppose A is independent of B , $P(A^c) = 0.6$ and $P(A \cup B) = 0.6$. What is the probability of $P(A \cap B)$? Explain.

Solution. $P(A) = 1 - 0.6 = 0.4$. Let $P(B) = p$. By independence, $P(A \cap B) = 0.4p$. Hence $0.6 = P(A \cup B) = 0.4 + p - 0.4p = 0.4 + 0.6p$, so $p = 1/3$. Therefore $P(A \cap B) = 0.4(1/3) = \boxed{\frac{2}{15} \approx 0.133}$.

8. Let A, B be arbitrary events. Show that $P(A) + P(B) = P(A \cup B) + P(A \cap B)$ from the three probability axioms.

Solution. Write $A = (A \setminus B) \cup (A \cap B)$ and $A \cup B = (A \setminus B) \cup B$, where both unions are disjoint. Thus $P(A) = P(A \setminus B) + P(A \cap B)$ and $P(A \cup B) = P(A \setminus B) + P(B)$. Subtracting the common term $P(A \setminus B)$ gives

$$P(A) + P(B) = P(A \cup B) + P(A \cap B).$$